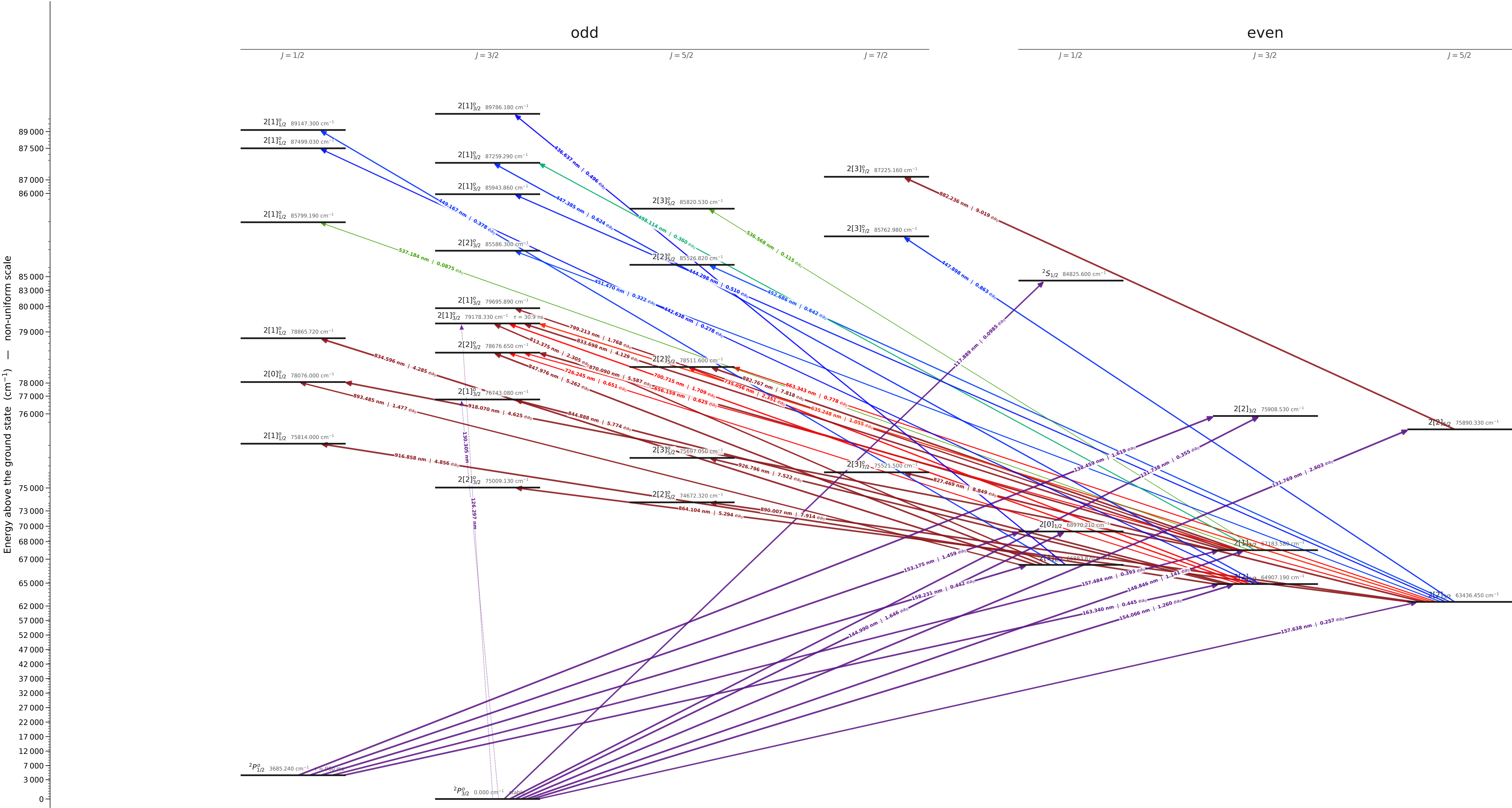
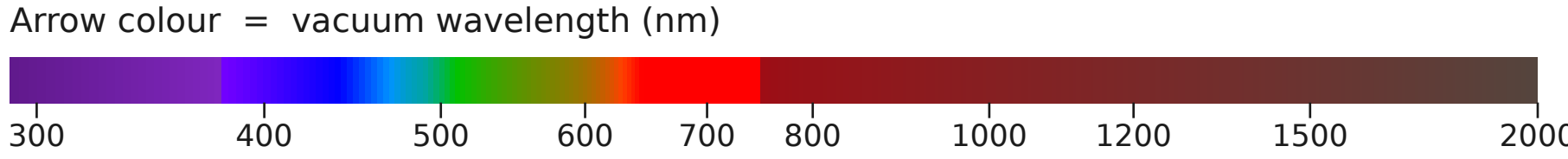


Bromine

Br (Br I). Strongest classified transitions below 2 μm (NIST): 33 levels, 47 lines.

Br⁺ ionisation limit 95284.80 cm⁻¹ = 11.81381 eV (104.949 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|U||e||J'J''|$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
² P _{3/2} - 2[1] _{3/2} ^o	126.2972	-	2373.7066	-	-	5.151	-	TALIF detection of ground-state Br: 2 x 252.594 nm, fluor...
² P _{3/2} ^o - 2[1] _{3/2} ^o	130.3049	-	2300.6997	-	-	-	-	two-photon LIF scheme near 260.6 nm with fluorescence Sp...
2[2] _{5/2} - 2[1] _{3/2} ^o	635.2481	635.0725	471.9297	1.055	2.200e+06	5.151	6.8	635.25 nm fluorescence line used to detect Br atoms after...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

⁷⁹Br hyperfine structure (I = 3/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
² P _{3/2}	884.80977(6)	-384.88284(20)	F=0: -3799.14 F=1: -2529.45 F=2: -374.94 F=3: +1894.60
² P _{1/2}	5332.55(4)	-	F=1: -6665.69 F=2: +3999.41

Measured A, B constants; sources in the data file.